

Pardeep Walia

Toronto, ON | 647-446-4867 | wpardeep007@gmail.com | [GitHub](#) | [LinkedIn](#) | [My Portfolio](#)

Professional Summary

Data Analyst & AI Engineer with strong expertise in SQL, Python, and Large Language Models (LLMs). Experienced in building agentic RAG workflows, self-healing data pipelines, and quantitative reporting systems. Proven ability to bridge the gap between complex data engineering and actionable business insights, utilizing tools like **LangGraph**, **Docker**, **Tableau**, and **DuckDB**. Committed to delivering high-accuracy, audit-ready solutions in fast-paced environments.

Technical Expertise

- **AI & GenAI:** RAG, LangChain, LangGraph, Ollama, Prompt Engineering, RAGAS Evaluation.
- **Deep Learning (Architectures):** Transformers (Attention Mechanisms), CNNs, RNNs, LSTMs.
- **Languages & Backend:** Python, SQL, NoSQL, FastAPI, REST APIs, Excel (Advanced Macros/VLOOKUP).
- **Data Engineering:** DuckDB, Docker (Compose, Containerization), Data Cleaning, ETL Pipelines, Anomaly Detection.
- **Frameworks & Libraries:** PyTorch, Pandas, NumPy, Pydantic, Scikit-learn.
- **Visualization:** Tableau, Power BI, Streamlit, Plotly.
- **Machine Learning:** Logistic/Linear Regression, SVMs, Decision Trees, Gradient Descent.

Professional Experience

Data Annotator | Cohere | Remote (Sep 2025 – Present)

- **Execute RLHF workflows**, identifying risks, hallucinations, and logic failures in AI model outputs.
- Analyze complex prompt-response pairs to improve model reasoning capabilities, directly contributing to model safety and alignment.

Data Analyst Intern | HelloFresh | Toronto, ON (Nov 2024 – Nov 2025)

- Performed **quantitative data validation**, identifying anomalies, correcting discrepancies, and ensuring accuracy before results were consumed by internal business teams.
- Used **SQL and Excel** to reconcile multi-source datasets, verify calculations, and investigate data integrity issues—mirroring first-level risk checks used in financial reporting.
- Produced recurring operational reports with strict turnaround times and contributed to **process automation** for data workflows.

Volunteer Data Analyst | Youth Research & Education Society (YRES) | Remote (Jan 2025 – Present)

- Produced ad-hoc Excel reports and cleaned multi-source datasets for quality and accuracy.
- Provided analytical summaries with validated datasets, applying a risk-style detail-oriented approach.

Key Projects

Agentic RAG Analytics Engine | *Python, LangGraph, Docker, DuckDB*

- Engineered a **local-first, self-healing analytics system** capable of ingesting arbitrary CSV datasets and executing NL-to-SQL queries without predefined schemas.
- **Achieved 0.91 faithfulness score** (via RAGAS) by implementing hallucination mitigation techniques, including schema grounding and SQL sanitization.
- Developed a resilient backend using **FastAPI** and **DuckDB**, featuring auto-retries for query failures and state management via **LangGraph**.
- Containerized the application using **Docker** to ensure environment-agnostic deployment and built an interactive frontend using **Streamlit**.

Job Market Data Analysis | *Python, SQL, Tableau*

- Built a large-scale dataset (200K+ rows) and applied structured SQL/Python transformations to compute metrics similar to portfolio exposure: distributions, segment breakdowns, variance, and trends.
- Conducted **first-level data checks** (range limits, missing values, threshold validations) before generating automated reports and dashboards.

Education & Certifications **York University:** Bachelor of Arts (Honours) Information Technology (2022 – 2025)

- *Relevant Coursework:* Data Analytics, Database Systems, Statistics, SAP ERP, UI/UX Design

Certifications:

- Tableau Desktop Specialist (Oct 2025)
- AWS Cloud Practitioner (Oct 2025)